

There are total 50 questions in this Test Booklet. Each question is having four alternative responses, of which one response is completely correct. Record the completely correct response on the OMR sheet by darkening the corresponding circle with Blue/Black ball point pen only.

1. Why can left recursion not be used in top-down parsing (like recursive descent parsers)?  
(A) It increases the complexity of the grammar  
(B) It creates ambiguity in the parse tree  
(C) It causes infinite recursion  
(D) It violates LL(1) property
2. LL(1) parsing uses one token lookahead. What does this limit imply?  
(A) It cannot parse ambiguous grammars (B) It can parse only left-recursive grammars  
(C) It can use more than one stack (D) It parses in quadratic time
3. In an LR parser, a shift-reduce conflict means:  
(A) A terminal is matched incorrectly  
(B) The parser cannot decide whether to reduce a production or read another symbol  
(C) Two productions match the same input  
(D) A non-terminal is incorrectly expanded
4. Which of the following is typically not checked during semantic analysis?  
(A) Type mismatches (B) Undeclared variables  
(C) Operator precedence (D) Function parameter consistency
5. What is the purpose of using intermediate code in a compiler?  
(A) To make lexical analysis faster (B) To enhance code optimization opportunities  
(C) To reduce source code dependencies (D) To allow execution without machine code
6. What does bootstrapping refer to in compiler construction?  
(A) Rewriting a compiler in machine code  
(B) Generating an interpreter from a compiler  
(C) Writing a compiler in the same language it compiles  
(D) Debugging a compiler using itself

7. Consider the following code fragment inside a loop:

```
for (int i = 0; i < n; i++) {
    int x = a * b;
    arr[i] = x + i;
}
```

Handwritten notes:  $arr[0] = 0$ ,  $arr[1] = 1$ ,  $arr[2] = 2$ ,  $arr[3] = 3$

(4)

Which of the following optimization strategies would result in reduced runtime overhead and is safe to perform here?

- (A) Convert x into a macro to eliminate runtime allocation.
- (B) Perform common subexpression elimination to merge x and i.
- (C) Apply loop-invariant code motion to move  $x = a * b$  outside the loop.
- (D) Inline the entire loop body to reduce branching.

8. What is the time complexity of Binary Search in the worst case?

- (A)  $O(n)$
- (B)  $O(n \log n)$
- (C)  $O(\log n)$
- (D)  $O(1)$

(4)

9. Overfitting in a machine learning model typically means:

- (A) The model is too simple and underperforms
- (B) The model performs well on training data but poorly on unseen data
- (C) The model has high bias and low variance
- (D) The model is well-generalized

(-1)

10. Which data structure is best suited for implementing a recursive function's call?

- (A) Queue
- (B) Array
- (C) Linked List
- (D) Stack

11. The recurrence relation  $T(n) = 2T(n/2) + n$  corresponds to which algorithm's time complexity?

- (A) Merge Sort  $O(n \log n)$
- (B) Linear Search  $O(n)$
- (C) Binary Search  $O(\log n)$
- (D) Bubble Sort  $O(n^2)$

12. In a paging system, which of the following is true about internal fragmentation?

- (A) Occurs between two allocated memory blocks
- (B) Happens when a process does not use the entire allocated page
- (C) Caused due to external memory blocks being too small
- (D) Arises due to variable page sizes

13. Which scheduling algorithm may cause starvation?

- (A) Round Robin
- (B) FCFS (First-Come, First-Served)
- (C) SJF (Shortest Job First)
- (D) Multilevel Queue with aging

14. The fork() system call in Unix creates:

- (A) A new kernel
- (B) A new thread
- (C) A new process with a new PID
- (D) A child thread in the parent process

15. A system uses paging with a page size of 4 KB (1 KB = 1024 bytes). A process generates a logical address of 15,385 bytes. What is the page number and the offset within that page?

- (A) Page number = 3, Offset = 3,097
- (B) Page number = 4, Offset = 1,289
- (C) Page number = 3, Offset = 1,289
- (D) Page number = 4, Offset = 3,097

16. Which attribute is used in HTML to open a link in a new tab or window?

- (A) href = "\_new"
- (B) target = "\_blank"
- (C) new = "tab"
- (D) link = "external"

17. What does the <form> tag in HTML do?

- (A) Creates a new frame
- (B) Defines an image map
- (C) Collects user input
- (D) Links CSS files

18. Which of the following is a server-side scripting language?

- (A) JavaScript
- (B) HTML
- (C) CSS
- (D) PHP

19. If  $A = \{a, b\}$ , what is the Cartesian product  $A \times A$ ?

- (A)  $\{(a, b)\}$
- (B)  $\{(a, b), (b, a)\}$
- (C)  $\{(a, a), (a, b), (b, a), (b, b)\}$
- (D)  $\{(a, b), (b, b)\}$

20. If  $A = \{1, 2, 3\}$ ,  $B = \{2, 3, 4\}$ , then  $(A \cup B) - (A \cap B)$  is:  
 (A)  $\{1, 2, 3, 4\}$  (B)  $\{1, 4\}$   
 (C)  $\{2, 3\}$  (D)  $\{2, 4\}$
21. How many different ways can the letters of the word "COTTON" be arranged?  
 (A) 720 (B) 60  
 (C) 120 (D) 180
22. If  $f(x) = 2x + 1$ , then  $f^{-1}(x)$  is:  
 (A)  $x-1$  (B)  $(x-1)/2$   
 (C)  $1/(2x-1)$  (D)  $(x+1)/2$
23. The mean of five numbers is 30. If one number is removed, the mean becomes 25. What is the value of the removed number?  
 (A) 30 (B) 25  
 (C) 50 (D) 100
24. A card is drawn from a standard 52-card deck. What is the probability of drawing a king or a heart?  
 (A)  $4/52$  (B)  $1/4$   
 (C)  $2/13$  (D)  $4/13$
25. Find the output of the following C code:  

```
#include<stdio.h>
int main(){
    int x;
    x=(1,2,3);
    printf("%d",x);
    return 0;
}
```

  
 (A) Compilation Error (B) 1  
 (C) 2 (D) 3

26. Find the output of the following C code:

```
#include <stdio.h>

int main() {
    int x=2;
    if(x=3)
        printf("Hi, the value of x=%d", x);
    else
        printf("Hello, the value of x=%d", x);
}
```

4

- (A) Hi, the value of x = 2  
(C) Hi, the value of x = 3

- (B) Hello, the value of x = 2  
(D) Compilation error

27. Find the output of the following C code:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[20] = "ABCDEFGHIJK";
    int s = strlen(str);
    str[3] = '\0';
    s = strlen(str);
    printf("%d\n", s);
    return 0;
}
```

4

1 2 3

- (A) 10  
(C) 3

- (B) 20  
(D) 4



28. What will be the output of the following C program?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, sum = 0;
```

```
    for (i = 1; i <= 5; i++) {
```

```
        if (i % 2 == 0)
```

```
            continue;
```

```
            sum += i;
```

```
    }
```

```
    printf("%d\n", sum);
```

```
    return 0;
```

```
}
```

(A) 9

(C) 5

(B) 15

(D) 6

$$i + 3 + 5 = 9$$

4

29. What will be the output of the following C program?

```
#include <stdio.h>
```

```
int add(int a, int b)
```

```
{
```

```
    return a + b;
```

```
}
```

```
int main()
```

```
{
```

```
    int x = 5, y = 10;
```

```
    printf("%d\n", add(x, add(x, y)));
```

```
    return 0;
```

```
}
```

(A) 20

(C) 25

(B) 15

(D) 30

$$+15 \\ 5 + 15$$

4

1  
16

1 1 0

9

4

1

4

1

30. A JK flip-flop with both  $J = 1$  and  $K = 1$  will:
- (A) Remain in the same state (B) Toggle its output  
(C) Reset the output (D) Set the output
31. If  $A = 1$  and  $B = 1$ , the output of a Half Adder is:
- (A) Sum = 1, Carry = 0 (B) Sum = 0, Carry = 1  
(C) Sum = 1, Carry = 1 (D) Sum = 0, Carry = 0
32. In which topology do all devices connect to a central hub or switch?
- (A) Mesh (B) Ring  
(C) Bus (D) Star
33. Which of the following is the loopback IP address in IPv4?
- (A) 127.0.0.1 (B) 0.0.0.0  
(C) 255.255.255.255 (D) 192.0.2.1
34. What is the main challenge associated with symmetric key cryptography?
- (A) Slower performance (B) Key distribution and management  
(C) Inability to sign messages (D) Lack of confidentiality
35. In asymmetric key cryptography, which key is used to encrypt the message?
- (A) The sender's private key (B) The receiver's public key  
(C) The receiver's private key (D) A shared secret key
36. Which of the following is not necessarily true about a foreign key?
- (A) It ensures referential integrity  
(B) It can reference a unique non-primary key  
(C) It must reference a column with a NOT NULL constraint  
(D) It may accept NULL values unless specified otherwise

37. If an ER diagram has a many-to-one relationship from *Employee* to *Department*, then:

- (A) A department can belong to one employee
- (B) Each employee belongs to exactly one department
- (C) An employee can belong to many departments
- (D) Each department must have at least one employee

*m* to *1*  
*Employee* to *Department*

38. What will the query `SELECT * FROM Employee WHERE Salary = NULL;` return?

- (A) All employees with NULL salary
- (B) Employees whose salaries are not set
- (C) Syntax error
- (D) No rows

39. Which statement about functional dependencies is incorrect?

- (A) If  $A \rightarrow B$  and  $B \rightarrow C$ , then  $A \rightarrow C$  (transitivity)
- (B)  $A \rightarrow A$  always holds
- (C) If  $A \rightarrow B$  and  $A \rightarrow C$ , then  $A \rightarrow BC$
- (D) If  $A \rightarrow B$ , then  $B$  is a candidate key

40. Which of the following statements best describes a limitation of the pumping lemma for regular languages?

- (A) It cannot prove a language is regular
- (B) It works only on finite languages
- (C) It can only be used to prove non-regularity, not regularity
- (D) It applies only to context-free languages

41. The Halting Problem is undecidable. Which implication follows from this?

- (A) Every problem that halts is undecidable
- (B) No algorithm can simulate another
- (C) There exists no Turing machine that can determine if another halts on all inputs
- (D) All problems in P are undecidable

42. Which language is not context-free?

- (A)  $L = \{a^n b^n \mid n \geq 0\}$
- (B)  $L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i = j \text{ or } j = k\}$
- (C)  $L = \{a^n b^n c^n \mid n \geq 1\}$
- (D)  $L = \{a^n b^m \mid n \neq m\}$



43. Suppose a DFA has  $k$  states. What is the maximum length of a string that must be checked to determine equivalence of two DFAs?
- (A)  $k$  (B)  $k^2$   
(C)  $2^k$  (D)  $k - 1$
44. Which of the following problems is undecidable?
- (A) Whether a DFA accepts all strings  
(B) Whether a CFG generates an infinite language  
(C) Whether a TM halts on a blank tape  
(D) Whether a regular expression describes an empty language
45. Which of the following is true about Turing Machines (TM) and Pushdown Automata (PDA)?
- (A) Every language accepted by a PDA is also accepted by a TM  
(B) PDAs are more powerful than TMs  
(C) Every regular language requires a TM with infinite tape  
(D) TMs can accept only recursive languages
46. Which of the following best describes an inherently ambiguous context-free language?
- (A) It has multiple grammars (B) Every grammar that generates it is ambiguous  
(C) It cannot be accepted by any PDA (D) Its parse trees are all unique
47. Which of the following is true for recursively enumerable (RE) languages?
- (A) They are always decidable (B) They are closed under complement  
(C) Every recursive language is RE (D) They can be accepted by finite automata
48. According to Rice's Theorem, which of the following is decidable?
- (A) Whether a TM halts on all inputs (B) Whether a TM accepts a finite language  
(C) Whether a TM accepts nothing (D) Whether a TM uses more than 1000 tape cells
49. Which transformation can cause exponential growth in state size?
- (A) NFA to DFA (B) DFA to CFG  
(C) CFG to PDA (D) TM to TM in normal form
50. Which of the following is not typically handled during lexical analysis?
- (A) Removal of comments (B) Syntax error detection  
(C) Identification of tokens (D) Skipping whitespace
- \* \* \* —